

“I’m Always Here”: An Affective Companion Agent for Novice Drivers

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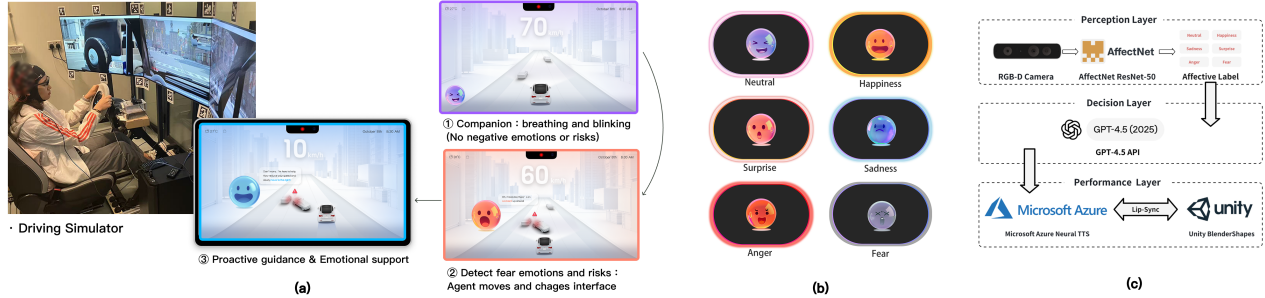


Figure 1: Overview of the Affective Companion Agent. (a) Interaction Flow. (b) Animation Interface. (c) System Architecture.

Abstract

Novice drivers face operational and emotional challenges but are often overlooked in current in-vehicle agent design. Based on the interviews with 22 novice drivers, we introduce an affective companion in-vehicle agent that monitors drivers’ expressions and proactively provides situational guidance and emotional support through animated interface and voice interaction without wake word. We developed a design prototype and implemented it in a driving simulator. The evaluation (N = 20) showed that users provided positive ratings for interaction quality, emotional support, and overall satisfaction, indicating the system’s potential for practical in-vehicle applications. This study provides new insights for the design of in-vehicle affective agents.

CCS Concepts

• **Human-centered computing** → **Human Computer Interaction**.

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1 Introduction

Novice drivers represent a notable portion of the driving population and face greater challenges compared to experienced drivers. Previous studies have often framed novices primarily as traffic risks from a critical perspective, overlooking their pain points, needs, and expectations for emotional support [McKnight and McKnight 2003]. At the early stage of driving, what novices most desire is a companion role akin to a “coach” or “co-driver”—one that not only provides guidance in complex situations but also offers reassurance and a sense of security when they feel anxious or stressed. While this role is often played by experienced family members or friends, interpersonal co-driving can easily lead to emotional friction. From this perspective, an intelligent system that is rational and equipped with extensive driving knowledge may represent a more effective solution.

With the increasing integration of affective computing and embodied AI into intelligent cockpits [Duan et al. 2022; Wang et al. 2022], affective in-vehicle agents have emerged as a promising approach to addressing both operational and emotional challenges faced by novice drivers. However, most existing agents are designed with experienced drivers in mind, assuming sufficient knowledge and skills, while the specific needs of novice drivers remain largely overlooked.

2 Formative Study

To understand novice drivers’ design expectations for in-vehicle affective systems, we conducted semi-structured interviews with 22 novice drivers. Through thematic analysis, we inductively generated four core themes and sixteen sub-themes.

The results indicated that they require the systems capable of perceiving both the driver’s state and road conditions. Since they

are often unable to accurately assess the correctness of their driving operations, such systems should accompany them throughout the entire driving process and be able to intervene proactively at critical moments. It is emphasized that emotion is regarded as a key requirement: novices express a need for positive and natural affective feedback. These findings led us to propose an affective companion-type in-vehicle agent. In contrast to voice assistants that must be explicitly activated, the affective companion agent remains persistently situated in the corner of the dashboard, intervenes proactively when it recognizes driving situations beyond the user's capacity to manage, and provides support through anthropomorphic user interface and voice interaction.

3 System Design

System Architecture. Based on the Sense-Think-Act model in affective computing [Dawson 2013; Picard 2000], the system comprises three layers: Perception, Decision, and Performance (Figure 1c). The Perception Layer uses an in-vehicle RGB-D camera and the pre-trained AffectNet ResNet-50 [Fard et al. 2024] model to classify real-time facial emotions into six categories [Ekman et al. 1999]: neutral, happiness, sadness, fear, surprise, and anger. The Decision Layer incorporates a rule engine to determine the timing of interventions and invokes the GPT-4.5 API as the semantic generation module, ensuring responses are natural and diverse within safe parameters. When neutral or positive emotions are detected, the agent remains in companion mode without voice interaction. If negative emotions and potential driving risks are identified, the agent adjusts its position and interface, generating driving guidance and emotionally supportive voice prompts. The Performance Layer, developed in Unity3D, controls the agent's animated expressions and movements, with speech synthesis powered by Microsoft Azure Neural TTS and synchronized lip movements for coherent affective feedback.

Interaction Flow. During normal driving, the affective companion agent remains unobtrusively in the corner of the dashboard. When stress, confusion, or unexpected hazards are detected, it transitions from a passive to an active state, displaying empathetic expressions and offering spoken guidance. To minimize distraction, visual prompts were limited to peripheral animation cues rather than detailed text during driving. For instance, when the driver becomes startled by a sudden obstacle, the agent moves to a central position, its facial expression softens, and it provides calm verbal instructions such as reducing speed and maintaining lane control. This process allows the agent to proactively guidance the driver through both situational and emotional support. Figure 1a illustrates this interaction flow.

User Interface. To correspond with the six emotion labels, we designed six affective agent animation interface (Figure 1b). The agent adopts an anthropomorphic design, with color schemes informed by color psychology and integrated ambient lighting that matches its affective state, supporting compatibility with standard HMIs. Since interview participants expressed a preference for "natural and continuous companionship", the interface incorporates micro-expressions such as breathing and blinking, enhancing both interactivity and the sense of presence.

4 Evaluation

We conducted a user study with 20 novice drivers using a driving simulator under various road conditions. Participants interacted with the affective companion agent while completing driving tasks. Satisfaction, interaction quality, and emotional support were assessed using a mixed-methods approach. As shown in Figure 2, users reported high ratings for interaction quality ($M = 4.19$) and emotional support ($M = 4.22$), highlighting the agent's effective companionship and timely guidance. Overall satisfaction ($M = 4.31$) indicated practical value. However, one participant felt the agent's animation reduced driving seriousness, suggesting the need to balance emotional support with driving rigor in future designs.

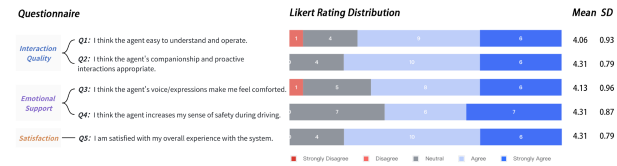


Figure 2: Statistical results of the evaluation questionnaire

5 Conclusion

Our study introduces a companion affective in-vehicle agent with an anthropomorphic animation interface and voice interaction. Based on interviews and thematic analysis with 22 novice drivers, we identified design expectations regarding affective companionship and proactive interaction, which informed the development of a prototype. We conducted a follow-up evaluation with 20 of the participants from the initial interview study. It is shown ($N = 20$) positive ratings for interaction, emotional support, and satisfaction, highlighting both application potential and the need for further refinement. While current results are promising, future work will focus on improving response timing and ensuring non-distractive affective feedback for safe real-world deployment.

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